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WHITE PAPER & FORENSIC INDUSTRY ANALYSIS · 2026

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The Knowledge Disease

How America's Small Trucking Companies Are Being Licensed to Fail — and the Circular System That Can Save Them

The current system licenses ignorance at the front door and bills it for the consequences at every door after.

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Prepared in support of the Heavy-Duty Repair Transparency & Fairness Act (HD-RTFA)

THE KNOWLEDGE DISEASE

A Forensic Analysis of the Structural Collapse of the American Small Carrier and Freight Brokerage — and the Circular Survival Imperative

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DOCUMENT TYPE	White Paper & Forensic Industry Analysis
DATE	July 2026
COMPANION TO	The Heavy-Duty Repair Transparency & Fairness Act (HD-RTFA)

This document is a neutral, evidence-driven industry analysis. Every major claim carries a named, dated source, and every significant evidence gap is disclosed. It is offered as evidence to legislators and regulators, as a systems analysis to industry and institutional partners, and as a plain-language explanation to the operators and citizens the freight system serves.

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Abstract

The United States small-carrier trucking sector and the independent freight brokerage that serves it are undergoing a severe, structural contraction. Between eighty-five and ninety percent of new owner-operator and small-carrier businesses fail within their first two years. This paper argues that the failures conventionally blamed on freight-rate cycles, insurance inflation, and equipment costs share a single upstream cause that the federal system licenses into existence: an epistemological deficit — a knowledge gap — created at the moment of entry. A person can obtain interstate operating authority for roughly a three-hundred-dollar filing fee with no proficiency examination, no financial-literacy screen, and no demonstration of operational competency, a knowledge gate that Congress twice ordered and the Federal Motor Carrier Safety Administration never built.

Drawing on 2026 data from the American Transportation Research Institute, the Federal Motor Carrier Safety Administration, the American Trucking Associations, the Government Accountability Office, insurers, and the courts — set beside more than two decades of the author's direct operational experience with small carriers and brokerages — this analysis identifies four proximate mechanisms of failure (repair and maintenance catastrophe, the insurance spiral, fraud and predatory intermediation, and regulatory compliance burden) and demonstrates that each becomes lethal only because of the knowledge deficit beneath it. Ignorance is the disease; the other four are how it kills.

The paper's central prescription is a paradigm shift from a linear, extractive operating doctrine to a circular logistics architecture — a regenerative model in which every operational decision is optimized for long-term survivability, and in which knowledge, capital, maintenance, compliance, and human performance are continuously reinvested back into the carrier. The evidence does not support a clean, datable extinction of the small carrier. It supports something more precise and more actionable: a staged deterioration and consolidation that will hollow out the independent middle of the industry within the next fifteen to twenty years unless the operating model and the competency gate are changed. This document is offered as evidence in support of that change, and as a companion to the Heavy-Duty Repair Transparency and Fairness Act.

A Note on This Document

This is a white paper and a forensic industry analysis. It is written to do three things at once. For a legislator or regulator, it is evidence — every major claim carries a named, dated source, and every significant gap in the evidence is disclosed rather than hidden. For a corporation, an insurer, a lender, or an institutional partner, it is a systems analysis of a market that most of the economy depends on and few outside it understand. And for the ordinary reader — the owner-operator, the new broker, the driver, or simply the American who eats food that arrived on a truck — it is a plain-language explanation of why good, hardworking people keep losing everything, and what can be done about it.

I have set my own experience beside the published data throughout. Where a figure is uncertain or contested, I have said so. I would rather under-claim and be right than over-claim and be dismissed, because this argument is too important to lose on a technicality. Read it as an operator, an official, a researcher, or a citizen — and then decide whether the system we have built is the one we meant to build.

— Michael Thomas, Founder, Freight University and Logistical Forwarding Solutions

SECTION 1

The Structural Contradiction

Why an industry the whole economy depends on keeps killing the businesses that run it

Begin with a fact that should end every argument about whether this matters. Trucks move 72.7 percent of America's domestic freight by tonnage — 11.27 billion tons in 2024 — and 76.9 percent of its freight revenue, roughly 906 billion dollars, according to the American Trucking Associations. Seven of every ten agricultural products in this country reach the table on a truck. When people say "the supply chain," a great deal of what they mean is a driver, a trailer, and a diesel engine.

Now look at who owns those trucks. According to FMCSA and ATA data, 91.5 percent of all registered carriers operate ten or fewer power units. Ninety-six percent operate fewer than twenty. Ninety-nine point three percent operate fewer than one hundred. Nearly half of all registered trucking entities are a single person and a single truck. The popular image of trucking as a business of giant fleets is precisely backwards. Trucking is a nation of small businesses — hundreds of thousands of them — that happen to carry the whole economy on their backs. The largest ten truckload carriers together control less than six percent of the market. The other ninety-four percent is the small carrier.

Here is the contradiction at the center of this paper. The U.S. trucking market is a near-perfectly deregulated commodity market layered on top of a heavily regulated safety regime. Since the Motor Carrier Act of 1980 deregulated economic entry, anyone may enter for the price of a filing fee. What was never deregulated is the safety-and-compliance stack: the Federal Motor Carrier Safety Regulations, the drug-and-alcohol testing program and its Clearinghouse, the electronic logging mandate, the fuel-tax and registration systems, and a dozen state permits. The economic barrier to entry is close to zero. The operational competence required to survive is enormous. And no one verifies, teaches, or gates that competence before granting authority.

The result is a churn machine. New trucking entities exploded from 72,566 registrations in 2018 to 281,376 in 2022. Roughly twice as many carriers now enter the industry each year as regulators can audit. Crashes involving these new carriers caused nearly five hundred fatalities in 2022, with an estimated 5.5-billion-dollar economic impact. This is not a market finding its equilibrium. It is a system that admits the unprepared at the front door and bills them — and the public — for the consequences at every door after.

SECTION 2

The Verified Deterioration

What the 2026 data actually shows

The failure is not anecdotal, and it is not new, but it has sharpened. The most rigorous industry estimate, from ATBS — the nation's largest owner-operator tax and accounting firm — is that eighty-five to ninety percent of new owner-operator businesses fail within the first two years, primarily from cash-flow problems and the underestimation of operating costs. Foley and Truckstop cite the same band; some sources widen it to ninety-five percent. In 2023, approximately 88,000 carriers exited — a closure rate near twelve percent against roughly 750,000 active carriers, the largest net loss in a quarter-century. Attrition continued through 2024 and moderated in 2025 to roughly 2,125 carriers leaving per month, even as the for-hire population remained about a third larger than before the pandemic.

The cost structure explains the pressure. ATRI's 2025 analysis of operational costs found the average marginal cost of operating a truck was 2.260 dollars per mile in 2024. With volatile fuel stripped out, the non-fuel marginal cost rose 3.6 percent to a record 1.779 dollars per mile — the highest figure ATRI has ever recorded. Truck and trailer payments jumped to a record 0.390 dollars per mile. Insurance premiums reached a record 0.102 dollars per mile. And the truckload sector — where the small carrier lives — ran an average operating margin of negative 2.3 percent. Empty, non-revenue "deadhead" miles rose to 16.7 percent, and drivers-per-truck fell to 0.93 as operators parked equipment they could not run at a profit.

Line item — ATRI 2025 (2024 data)	Per mile
Total marginal cost	**$2.260**
Non-fuel marginal cost	**$1.779 (record high)**
Fuel	**$0.481**
Truck & trailer payments	**$0.390 (record high)**
Repair & maintenance	**$0.198**
Insurance premiums	**$0.102 (record high)**
Driver wages	**~$0.799**
Empty (deadhead) miles	**16.7%**
Truckload operating margin	**−2.3%**

Behind the averages sits a fault line. Small carriers are overwhelmingly exposed to the spot market — the volatile, load-by-load pricing tier — while large fleets run on contract freight locked in months ahead. The same market that shows a negative average margin is one in which the largest fleets protect their rates while the smallest sell into the lower, more violent

spot tier and absorb every downturn. When spot rates recovered in early 2026, the recovery was driven by carriers exiting, not by demand rising. As one senior analyst put it, small carriers continuing to exit under sustained cost pressure "is not what a demand-based recovery looks like."

SECTION 3

The Three-Hundred-Dollar Door

A licensing anomaly with no equal in American commerce

To operate an eighty-thousand-pound vehicle in interstate commerce, hauling other people's freight for hire across state lines, an applicant must complete a wire transfer. That is not rhetoric. The USDOT number is free. Motor Carrier operating authority costs three hundred dollars per authority type, non-refundable. Activation requires a process-agent filing, a modest annual registration, and proof of insurance. That is the entire competency gate: money and paperwork.

Nowhere in this process does anyone verify that the applicant can calculate a cost per mile, read a rate confirmation, build a driver-qualification file, schedule preventive maintenance, fund a repair reserve, or recognize a fraudulent broker. There is no proficiency examination, no financial-literacy screen, and no demonstration of operational experience. The contrast with every other licensed field in the country is stark — and it is the single clearest way to see how strange this arrangement is.

Occupation	Competency requirement	Examination
Interstate motor carrier	None (post-entry safety audit only)	None
General contractor (CA)	4 years journey-level experience	Law/business + trade exam
Real-estate agent (CA / TX)	135 / 180 classroom hours	State licensing exam
Certified public accountant	150 semester hours + experience	Uniform CPA Exam
Licensed cosmetologist	~1,000–2,100 supervised hours	Written + practical board

A person who wants to legally cut hair must complete on the order of one thousand to twenty-one hundred supervised hours and pass a two-part state board examination. A person who wants to sell a house must complete a hundred and thirty-five classroom hours in California or a hundred and eighty in Texas and pass a licensing exam. A person who wants to certify financial statements must earn a hundred and fifty semester hours of college credit and pass one of the hardest professional exams in the country. A person who wants to operate a fleet of Class 8 trucks on public highways must complete a three-hundred-dollar transaction.

This is not an oversight that no one noticed. Congress noticed twice. Section 210(b) of the Motor Carrier Safety Improvement Act of 1999 directed the Department of Transportation to ensure that applicant carriers are knowledgeable about federal safety requirements before receiving new-entrant registration, and to consider a proficiency examination. In the resulting rulemaking, the agency, in its own words, "considered, but decided against requiring a

proficiency examination." In 2012, the MAP-21 reauthorization went further and mandated proficiency examinations. The exam was never built. As the trade press summarized in March 2026, FMCSA "never implemented the exam," relying instead on a safety audit during a carrier's first year — an audit that, by the agency's own data, is completed on time for barely half of new entrants. The last substantive action on the new-entrant knowledge rulemaking was an advance notice in 2009; a second advance notice is now projected, a regulatory stall approaching two decades.

The New Entrant Safety Assurance Program is real, and it is not nothing — but it is important to be precise about what it does. Usually within the first twelve months, it audits the carrier's driver-qualification files, drug-and-alcohol program, Clearinghouse queries, hours-of-service records, maintenance program, insurance, and accident register. Sixteen specific violations cause automatic failure. It is, in the words of one compliance firm, "a document inspection, not a road test. Most carriers who fail aren't unsafe — they're disorganized." Critically, it does not examine business viability, rate math, capitalization, or hiring controls. It verifies that paperwork exists, not that the operator understands the business. Roughly thirty to forty percent of new carriers fail the initial audit on the first attempt; civil penalties can reach 16,550 dollars per violation. These are not failures of malice. They are failures of knowing.

SECTION 4

The Five Killers, Ranked

How the deficit becomes a death

This analysis has evolved across three bodies of research. The first established the structural deterioration. The second corrected an error in the ranking of causes — it moved repair and maintenance from a "moderate" concern to the number-one proximate killer. The third, presented here, identifies the root beneath all of them. The result is a diagnostic ranking that distinguishes what strikes the final blow from what set the victim up to die.

1. Repair, maintenance, and downtime catastrophe — the proximate killer. The blade that severs the artery.
2. The insurance spiral — the chronic disease that weakens the patient for years before any acute event.
3. The fraud pandemic and predatory or incompetent brokers — the acute infection that a healthy operator survives and a fragile one does not.
4. The regulatory and compliance burden — the tax that knowledge renders survivable and ignorance turns into a death sentence.
5. The knowledge gap — the upstream root cause; the immune deficiency that makes every other killer lethal.

The organizing principle is simple. An educated operator survives a fifteen-thousand-dollar in-frame engine rebuild because they carried a reserve and knew their true cost per mile. An ignorant one is destroyed by the identical event. The knowledge gap never appears on a death certificate — "repair," "insurance non-payment," "authority revocation," or "fraud loss" does — but it is the comorbidity underlying all of them. The next five sections take the killers in the order that a carrier actually experiences them.

SECTION 5

The Repair Extraction Economy

The proximate killer, and the evidence behind the Heavy-Duty Repair Transparency and Fairness Act

All trucks break. That is not the problem. The problem is that the modern Class 8 truck has been engineered, financed, and serviced in a way that converts an ordinary mechanical failure into a business-ending event for the operator least able to absorb it — and that the profit structure of the manufacturers is deliberately aligned against the carrier's uptime.

The evidence for that last claim is not conspiracy; it is disclosed strategy. PACCAR, the largest publicly traded pure-play truck maker and parent of Kenworth and Peterbilt, reported a truck-segment gross margin of 13.9 percent in 2024 against an aftermarket parts gross margin near thirty percent — roughly 2.2 times higher. In the first quarter of 2026, PACCAR's Parts division out-earned its entire Truck division on a pretax basis by more than two to one. Rush Enterprises, the nation's largest truck dealer, posted an "absorption ratio" of 130.8 percent, meaning its parts-and-service gross profit more than covered the dealership's entire overhead before it sold a single truck. Volvo, Daimler, and the other majors show the same pattern in their own investor materials: the durable, recession-resistant profit is in keeping the installed base running, not in the sale. The industry calls this a "razor-and-blades" model. For the small carrier, it means the entire dealer-parts-service complex is optimized to maximize the spend on every repair event.

The real-world numbers are brutal and well documented. Independent heavy-duty shop labor reached a median of 149 dollars per hour in 2025; dealers charge more. An in-frame engine rebuild runs twenty to thirty thousand dollars, an out-of-frame replacement forty to fifty thousand. A full aftertreatment failure — the post-2010 emissions hardware of diesel particulate filter, selective-catalytic-reduction system, and diesel-exhaust-fluid dosing — can run twelve to twenty-three thousand dollars and is frequently excluded from warranty. Cooling-system and fuel-system work commonly runs into five figures. And ATRI's research on predatory towing found the average non-predatory heavy tow invoice was 8,925 dollars and the average predatory one was 18,154 dollars, with two-thirds of tows over thirty thousand dollars judged predatory.

Three structural traps make this worse for the smallest operators. The first is the emissions-failure tax: post-2010 systems fault, force a "derate" that can crawl a truck to five miles per hour, and require dealer-only diagnostics the owner cannot clear himself. The federal government itself conceded the burden in 2025 and 2026 by loosening derate rules, with the EPA acknowledging the existing system was "unacceptable" and cost small businesses millions in lost productivity. The second is the used-truck warranty cliff: small carriers buy three-to-four-year-old trucks precisely as the factory warranty expires and the major-component failure curve turns upward, and many financed at the 2021–2022 price peak,

leaving them underwater on the note while the largest repair bills arrive. The third is the downtime multiplier: unplanned downtime costs roughly 448 to 760 dollars per truck per day, so a single failure stacks the repair, the tow, one to two weeks of lost revenue, and often a lost customer into one event.

> *Put the pieces together and the arithmetic is fatal. A single major failure — repair, tow, a week or two out of service, and a missed load — routinely totals seventeen to twenty-eight thousand dollars against an owner-operator who nets fifty to sixty-five thousand dollars in a good year. One event consumes four to six months of income and, without a reserve, begins the documented death spiral: missed truck payment, damaged credit, higher insurance at renewal, worse factoring terms, and closure.*

This is the section of the analysis that most directly grounds the Heavy-Duty Repair Transparency and Fairness Act, the reform advanced by Logistical Forwarding Solutions and the author. The repair extraction economy is not a market that will correct itself, because the information asymmetry is the product. Right-to-repair access for heavy trucks remains gated behind proprietary manufacturer telematics; the federal REPAIR Act advanced in 2026 but had its strongest data-access provisions stripped in committee. The case for legislated repair transparency and fairness — honest diagnostics, honest parts pricing, honest access — rests on exactly the evidence above: the operator cannot survive a market whose costs are deliberately opaque and whose profit depends on that opacity.

SECTION 6

The Insurance Spiral

The chronic disease

If repair is the blade, insurance is the slow illness that leaves the patient too weak to survive it. Commercial auto liability has been unprofitable for insurers for fourteen consecutive years. ATRI put the cost at a record 0.102 dollars per mile in 2024, up roughly thirty-eight percent since 2015 — and this during a period when heavy-truck crash rates actually fell 2.6 percent. Premiums are not rising because the roads got more dangerous. They are rising because the courtroom did.

The driver is verdict severity, not accident frequency — the phenomenon insurers call "social inflation." The median nuclear verdict in trucking reached fifty-one million dollars in 2024, up from forty-four million in 2023 and twenty-one million in 2020. "Thermonuclear" verdicts above one hundred million dollars surged in 2024. Third-party litigation financing — private capital that funds lawsuits against carriers as an investment class — has grown into an eighteen-billion-dollar-a-year industry. Because every underwriter must now price the statistical possibility of a catastrophic verdict into every policy, the safe operator subsidizes the litigious environment, and the smallest fleets, with the least negotiating power, pay more than three times as much per mile for insurance as the largest ones. The federal minimum liability requirement of 750,000 dollars has not moved since 1985; against a fifty-one-million-dollar median verdict it is, in practice, a starting point for a negotiation that can end in the owner's personal assets.

The knowledge linkage is direct. An educated operator who runs cameras and telematics, keeps clean safety scores, and documents a real hiring standard reaches the preferred insurance tiers. The ignorant operator — who hires off the street and lets safety violations accumulate — becomes uninsurable, and an uninsurable carrier is simply a former carrier. Insurance is the chronic condition; knowledge is the difference between managing it and dying of it.

SECTION 7

Fraud, Chameleons, and the Broker Failure

The acute infection

The same zero-barrier door that admits the unprepared admits the criminal. A "chameleon" or reincarnated carrier shuts down under a record of violations and re-registers under a fresh identity; the modern variant clones the identity of a legitimate carrier to steal loads or double-broker freight. The Government Accountability Office documented that applicants with chameleon attributes were roughly three times as likely to be involved in severe crashes — eighteen percent versus six percent — and tied reincarnating carriers to 217 deaths and 3,561 injuries over a five-year window.

The financial fraud has since exploded. Strategic cargo theft — theft by deception rather than by force — rose 1,475 percent between 2022 and 2024 as overall cargo theft rose 93 percent. Confirmed losses reached roughly 725 million dollars in 2025, with the average theft valued at 273,990 dollars. Double-brokering, in which a fraudulent actor books a load and illegally re-brokers it to an unsuspecting carrier who moves the freight and is never paid, is now the most frequently reported freight-fraud scheme. One load board reported blocking more than nine hundred thousand fraudulent inbound emails in a single year. Legitimate MC numbers, surety bonds, and insurance certificates are all spoofable; the FMCSA itself warns that even insurance certificates can be fraudulent.

For the small carrier there is effectively no recourse. A single unpaid three-thousand-dollar load — a loss the author has personally absorbed through a "verified" counterparty on a major load board — severs the cash flow of a micro-carrier and can begin the same default sequence as a blown engine. The brokerage side of the industry suffers the mirror image of the carrier's disease: broker authority requires a seventy-five-thousand-dollar bond and a three-hundred-dollar fee, and no training whatsoever. Over eight thousand brokerages reportedly closed in 2023. The untrained broker who onboards a carrier sight-unseen, never verifying equipment or safety history, loses the shipper the day that carrier's aging truck strands the load on the shoulder — and loses the business with it. Fraud is the acute infection; the knowledge to prevent it, and the tools to detect it, are a few dollars and a written protocol away.

SECTION 8

Virtual Trucking and the Driver Carousel

The root cause, seen in the field

I have watched the knowledge gap kill more companies than any single mechanical failure, and it wears a recognizable face. I call it virtual trucking: the absentee entrepreneur who spends a hundred to a hundred and fifty thousand dollars to start a fleet, scales fast by onboarding drivers sight-unseen — one driver brings another, and another — never physically inspects the trucks, never meets the drivers, and is destroyed within a year. This is not a caricature. The startup capital figure is corroborated by multiple 2025–2026 industry guides; new-authority insurance alone runs fourteen to thirty thousand dollars per truck per year. What follows is documented mechanism, not misfortune.

I worked with a carrier that scaled from one truck to more than twenty in under seven months and was out of business roughly six months after that, destroyed by accumulated safety violations. I have been contacted by owners of large fleets who had never physically seen their own trucks; one discovered that all seven of his units needed fifteen to twenty thousand dollars each in neglected repairs — well over a hundred thousand dollars — and closed at a total loss near four hundred thousand. Drivers who do not care throw trucks to the ground: maintenance abuse, logging violations, safety-score destruction, and, in the worst cases, unvetted drivers in fatal crashes.

The compliance mechanics make the small fleet uniquely fragile here. The federal safety-scoring system groups carriers by inspection volume and ranks them by percentile. Because a small fleet generates very few inspections, a single bad driver's violations can spike its percentile past the intervention threshold in a matter of weeks. One bad hire, in a small fleet, is a company-ending event — and because turnover at large truckload carriers runs ninety to ninety-five percent, violation-prone drivers migrate endlessly from one failing carrier to the next, a carousel a desperate new operator hires from without knowing what he is buying.

The predictive data is unambiguous. ATRI's crash-prediction research, built on more than half a million driver records, found that a prior reckless-driving violation raises future crash likelihood by roughly a hundred to a hundred and fourteen percent; a prior crash, by a hundred and thirteen percent. Its litigation research found that improper hiring and onboarding practices raised expected jury awards by 272 percent. The landmark billion-dollar verdict of 2021 turned on a driver operating without a valid commercial license whom the carrier "never even bothered to check." And the tools that stop all of this are cheap and voluntary: the Pre-Employment Screening Program costs ten dollars a report and, by FMCSA's own study, cuts crash rates about eight percent and driver out-of-service rates seventeen percent. The safeguards cost a few dollars. The knowledge to use them is the entire gap.

SECTION 9

The Human Toll

What the numbers do to people

Beneath the ledgers is a human system in degradation, and it must not be read as a separate chapter from the financial one — it is the same chapter, read from the cab. The ninety-to-ninety-five-percent turnover treadmill reflects real biological and psychological wear: chronic fatigue, disrupted sleep, metabolic disease, isolation, and a pay structure that rewards miles instead of time. Roughly sixty-nine percent of long-haul drivers are obese and about half smoke — figures roughly double the general working population. Peer-reviewed work finds depression at twice the general rate, with severe depression associated with a 2.4-fold increase in collision risk. Between 2020 and late 2024 the federal Drug and Alcohol Clearinghouse recorded more than three hundred thousand violations; roughly two hundred thousand commercial drivers now sit in prohibited status.

When freight rates crash and the only lever an operator controls is output, drivers push their biological limits to make the math work — the documented "catch-up mentality" that produces hours-of-service violations, fatigue, and crashes. The unvetted driver, hired by a carrier too undercapitalized to be selective, is at once a victim of the system and a vector for its failures. A crash is a person's death, a driver's ruin, and a carrier's end, all in a single instant. The linear model treats human capital as a disposable input and extracts labor until the operator physically or legally breaks. A circular model cannot afford to, because in a circular model the driver's health is an asset the business depends on next month.

SECTION 10

The Circular Failure Model

Why education sits at the center of every loop

The five killers are not independent line items on a spreadsheet. They are one interlocking machine, and the knowledge gap is the central hub that turns every gear. Trace the loops and every one of them originates at, or passes through, that hub.

- Ignorance of true startup cost and reserve requirements produces undercapitalization, which turns a survivable repair into a fatal one.
- Ignorance of hiring discipline — no screening, no qualification files — produces safety-score destruction, which produces insurance spikes and non-renewal, which parks the trucks and ends the revenue.
- Ignorance of cost-per-mile produces hauling below cost, which starves cash, which defers maintenance, which produces downtime and violations, which tighten the loop again.
- Ignorance of fraud protocol, on the broker side, produces a double-brokered loss or a stranded load, which loses the shipper and ends the business.
- Ignorance of the compliance stack produces New Entrant audit failure, which produces authority revocation before the company ever really began.

This is why the knowledge gap is not merely the fifth killer but the root killer: it is the only variable that touches all four proximate mechanisms at once. Remove the hub — inject knowledge — and every loop loosens simultaneously. That single observation is the intellectual foundation of everything Freight University builds, and of the survival framework that follows.

SECTION 11

The Circular Logistics Survival Framework

Turning vicious loops into regenerative ones

If failure is a set of vicious loops, survival is the deliberate construction of balancing loops — the discipline of closing each knowledge gap before it compounds. Circular logistics is not a slogan and not an environmental posture. It is a systems doctrine: optimize every operational decision for the survivability of the next cycle, so that the revenue and knowledge from one loop fund the maintenance, compliance, and human performance of the next. Instead of asking "how do we make the most on this load," the circular operator asks "what decision today raises the probability that this company is still alive in five years." The framework has seven load-bearing practices.

1. Know your true cost per mile before accepting a single load. Against a 2024 marginal cost between 1.78 and 2.26 dollars per mile, an operator who cannot compute their own number is mathematically guaranteed to haul below cost eventually.
2. Operate a written driver onboarding and vetting standard — a Pre-Employment Screening Program query, a Clearinghouse check, a motor-vehicle record, employment verification, and a complete qualification file — for every driver, every time.
3. Fund and schedule preventive maintenance, enforce pre-trip inspections, and treat the warranty cliff as a known event rather than a surprise. Carry a repair reserve of at least one catastrophic event per truck before scaling.
4. Run a compliance calendar covering the full federal and state stack — the safety regulations, drug and alcohol testing and Clearinghouse queries, electronic logging and hours of service, fuel-tax and registration filings, and the biennial update.
5. Practice factoring and credit discipline: verify a broker's bond and payment history before hauling, and never move freight for an entity you have not vetted.
6. On the brokerage side, run a layered fraud protocol — carrier-vetting platforms, identity verification, and phone confirmation through the federal safety system before tendering any load.
7. Treat insurability as strategy, not overhead: cameras, telematics, clean safety scores, and a documented safety culture are what move a carrier from the assigned-risk pool into the preferred market.

Two structural moves sit above the individual practices. The first is the cooperative capacity network: independent operators, pooled into alliances, achieve the regional density that lets them present to enterprise shippers as a single large-scale provider, bypass predatory intermediaries, and command the purchasing power otherwise reserved for mega-fleets. The second is the eradication of deadhead through reverse logistics: with empty miles at 16.7 percent and retail returns exceeding eight hundred billion dollars a year, the operator who

turns a last-mile drop-off into a first-mile pickup converts pure loss into regenerative revenue. Each of these is a balancing loop. Together they are the difference between a straight line that ends and a circle that sustains.

SECTION 12

Policy: The Competency Gate and the Case for Reform

What Congress ordered, and what still must be built

Two reforms follow directly from this analysis, and Logistical Forwarding Solutions is advancing both. The first is repair transparency and fairness — the Heavy-Duty Repair Transparency and Fairness Act — grounded in the extraction economy documented in Section 5. A market whose profit depends on opaque diagnostics, opaque parts pricing, and gated access cannot be survived by the small operator; honest access to the truck's own data and honest pricing are the minimum conditions of a fair repair market. The second is broker transparency: the long-delayed modernization of the federal transaction-record rule, which would require electronic disclosure of transaction records within forty-eight hours and void the waiver clauses that let large brokerages hide their margins. Transparency is the bedrock of circular optimization — an operator cannot optimize a loop whose financial data is deliberately obscured by an intermediary.

But the deepest reform is the one Congress already ordered and never delivered: a competency gate at the front door. The intent of the 1999 and 2012 statutes was correct. A person about to bet a house and put an eighty-thousand-pound vehicle on a public highway should have to demonstrate, before receiving authority, a basic command of cost-per-mile literacy, an onboarding and driver-qualification standard, a maintenance-reserve plan, and the safety regulations that govern the work. This is not a barrier designed to keep people out. It is the same protection every other licensed profession already provides — and, applied here, it is the only intervention that reaches the population most at risk, because the absentee "virtual trucker" is precisely the operator least likely to seek training voluntarily.

Where the gate cannot yet be legislated, it can be built privately. The evidence that education changes survival is real, if not trucking-specific: mentored small businesses survive at markedly higher rates — one national program reports eighty-seven percent one-year survival with a mentor versus seventy-five percent without; another reports seventy percent five-year survival with mentoring, roughly double the unmentored rate. Freight University exists to be that gate — the voluntary equivalent of the competency requirement the federal system has failed to enforce for a quarter-century.

SECTION 13

Scenarios and Verdict

What the evidence supports, and what it does not

Honesty about the forecast is what makes this document usable as evidence, so let me be exact about what the data does and does not prove. It does not support a clean, datable extinction — the claim that every small trucking company will be gone in fifteen or twenty years. Low entry barriers guarantee that the class regenerates at the bottom even as it is culled, so the correct verdict is not extinction but staged deterioration and consolidation: perpetual churn at the bottom, and a progressive hollowing-out of the durable middle-tier operators whose failures destroy capacity, terminals, and skilled jobs that do not grow back.

Three scenarios follow from the current trajectory. Under the base case — no intervention — the eighty-five-to-ninety-percent failure rate persists, the front door keeps admitting both the unprepared and the evasive, and the churn machine runs on. Under a moderate case, a voluntary education and mentorship ecosystem expands and measurably lifts survival among those who seek it, while leaving the highest-risk absentee operators untouched. Under the structural case, the competency gate Congress ordered is finally built as a genuine business-and-safety requirement, and for the first time the intervention reaches the population that needs it most, because it operates at the gate rather than by opt-in.

> *The verdict, stated plainly: the thesis that the small carrier is structurally incompatible with the linear, extractive model is supported by the evidence. The knowledge gap is the disease; repair catastrophe, the insurance spiral, fraud, and regulatory failure are how it kills; and the three-hundred-dollar door is the epidemiological source. Until the competency gate is built, or the industry builds a voluntary equivalent that reaches the absentee operator, the failure rate is not a market inefficiency to be lamented. It is the predictable, designed output of a system that licenses ignorance and then bills it for the consequences. That system was not handed down by nature. We built it — and we can build a circular one in its place.*

Appendix A — Twenty-Five Key Findings

01 Motor Carrier operating authority costs three hundred dollars per authority type; the USDOT number is free (FMCSA).

02 No education, financial-literacy, or competency requirement precedes authority — only money and paperwork.

03 Congress ordered a knowledge gate twice — the Motor Carrier Safety Improvement Act of 1999 and MAP-21 in 2012 — and FMCSA never implemented the proficiency exam.

04 FMCSA relies instead on a post-entry New Entrant Safety Audit; roughly twice as many carriers enter yearly as can be audited, and barely half of audits finish on time.

05 Eighty-five to ninety percent of new owner-operator and small-carrier businesses fail within two years (ATBS via AtoB; Foley; Truckstop).

06 Approximately 88,000 carriers exited in 2023 — a roughly twelve-percent closure rate; about 2,125 carriers per month left through late 2025.

07 New trucking entities grew from 72,566 in 2018 to 281,376 in 2022; new-carrier crashes caused nearly 500 fatalities in 2022.

08 ATRI 2024 costs: 2.26 dollars per mile all-in; a record 1.779 dollars per mile non-fuel; the truckload operating margin was negative 2.3 percent.

09 Trucking moved 72.7 percent of domestic freight tonnage and 76.9 percent of freight revenue — some 906 billion dollars — in 2024 (ATA).

10 Contractors, real-estate agents, CPAs, and cosmetologists all require documented hours and licensing exams; interstate trucking requires none.

11 The New Entrant audit checks paperwork — qualification files, drug program, hours of service, maintenance, insurance — not business viability or rate math.

12 Roughly thirty to forty percent of carriers fail the initial safety audit on the first attempt; civil penalties can reach 16,550 dollars per violation in 2026.

13 Operator testimony: a carrier grew from one to more than twenty trucks in under seven months, then closed about six months later from accumulated safety violations.

14 Operator testimony: two fleet owners had never seen their own trucks; one found all seven units needed fifteen to twenty thousand dollars each in repairs, closing at a roughly four-hundred-thousand-dollar loss.

15 Safety-score percentiles are volatile for small fleets; one bad driver can breach the intervention threshold in weeks.

16 Large-truckload driver turnover runs ninety to ninety-five percent annually (ATA).

17 ATRI crash-predictor: a reckless-driving violation raises future-crash likelihood by roughly 104 to 114 percent; a prior crash, by 113 percent.

18 ATRI litigation research: improper hiring or onboarding raised expected jury awards by 272 percent; substance abuse, by 341 percent.

19 The 2021 billion-dollar Florida verdict turned on a driver operating without a valid commercial license whom the carrier never checked.

20 The Pre-Employment Screening Program costs ten dollars a report and cut crash rates about eight percent and driver out-of-service rates seventeen percent (FMCSA) — yet remains voluntary.

21 Broker authority requires a seventy-five-thousand-dollar bond plus the three-hundred-dollar fee and no training; more than 8,000 brokerages reportedly closed in 2023.

22 Strategic cargo theft rose 1,475 percent from 2022 to 2024; confirmed 2025 losses reached roughly 725 million dollars, averaging 273,990 dollars per theft.

23 GAO found chameleon-attribute applicants had eighteen-percent severe-crash involvement versus six percent, tied to 217 deaths and 3,561 injuries (2005–2010).

24 Commercial auto liability has been unprofitable for fourteen consecutive years; insurance reached a record 0.102 dollars per mile in 2024, up about thirty-eight percent since 2015.

25 The median nuclear verdict reached fifty-one million dollars in 2024, up from forty-four million in 2023 and twenty-one million in 2020; the federal 750,000-dollar minimum has not changed since 1985.

Appendix B — Data Gaps and Methodological Disclosure

Credibility requires disclosing the limits of the evidence as plainly as the evidence itself. The following qualifications apply.

- No federal dataset codes "knowledge gap" or "ignorance" as a cause of business closure. The eighty-five-to-ninety-percent failure figure is a robust industry consensus, not a single peer-reviewed federal statistic, and it blends owner-operator and small-carrier populations.
- New Entrant audit failure rates conflict across sources — roughly thirty to forty percent first-attempt failure per compliance vendors versus a high eventual pass rate per FMCSA framing. Both can be true (first attempt versus post-correction); the exact federal figure was not isolated here.
- Fraud, turnover, and blocked-attempt figures rely partly on vendor reporting with commercial incentives, though they are directionally corroborated across independent industry, insurer, and trade sources.
- The operator testimony in Section 8 is primary and firsthand. The specific dollar figures are corroborated in mechanism by independent cost ranges and safety-score dynamics, but the individual cases were not independently audited and are presented as such.
- The mentorship-survival evidence is economy-wide, not trucking-specific; no controlled study isolates the effect of pre-authority education on trucking survival. The balancing-loop case is built honestly on analogy.
- The MAP-21 proficiency-exam provision attaches most cleanly to household-goods carriers and brokers; the broader carrier-knowledge directive rests most cleanly on the 1999 statute. Both are cited to avoid overstatement.
- Certain 2026-dated references are forward-looking or projected — notably the projected second advance notice of rulemaking on new-entrant knowledge and pending regulatory actions — and are flagged as pending rather than settled.
- **On the forecast itself:** this analysis explicitly rejects a fixed "fifteen-to-twenty-year extinction" as unprovable, and adopts the more defensible verdict of staged deterioration and consolidation supported by current trends.

Appendix C — Sources and Classification

Sources are classified by evidentiary weight: A denotes government or primary regulatory sources; B, peer-reviewed research or major research institutes such as ATRI, the GAO, and the ATA; C, established trade press and legal-industry analysis; D, commercial or vendor data with a stated basis; and E, primary operator testimony, firsthand but not independently audited except where corroborated in the text.

Claim	Source	Class
$300 MC authority fee; free USDOT number; BOC-3, UCR, insurance filings	FMCSA registration	A
1999 & 2012 proficiency-exam mandates never implemented	FMCSA Final Rule (2008); HDT/TruckingInfo (3/2026)	A / C
New Entrant audit scope; 16 automatic-failure violations	FMCSA; 49 CFR Part 385	A
~30–40% initial audit failure on first attempt	Compliance vendors citing FMCSA	D
Costs $2.26 / $1.779 per mile; −2.3% truckload margin; 16.7% deadhead	ATRI 2025 Operational Costs report	B
85–90% of new owner-operators fail within two years	ATBS via AtoB; Foley; Truckstop	D
~88,000 carriers exited 2023; ~2,125/month leaving late 2025	AtoB; Motive; FTR	D
New entities 72,566 (2018) → 281,376 (2022); ~500 fatalities 2022	FMCSA data via industry analysis	A / C
Trucking moved 72.7% of tonnage, $906B revenue (2024)	American Trucking Associations	B
CSA/SMS percentile volatility for small fleets; intervention thresholds	FMCSA SMS Methodology; Forward Thinking Systems	A / C
Large-truckload turnover 90–95% annually	ATA via CCJ; OOIDA ‘The Churn’	B
Crash predictor: reckless +~104–114%; prior crash +113%	ATRI 2022 (580,000+ records)	B
Improper hiring/onboarding raised expected jury awards +272%	ATRI litigation research via CCJ	B
2021 $1B Florida verdict turned on an unlicensed driver never checked	Court records; legal-industry analysis	C
PSP costs $10/report; cut crash rates ~8%, driver OOS ~17%	FMCSA study	A
$75,000 BMC-84 broker bond; 8,000+ brokerages closed 2023	FMCSA; surety-industry data	A / D
Strategic theft +1,475% (2022–24); ~$725M losses, $273,990 avg (2025)	Verisk CargoNet; Truckstop; Highway	C / D
Chameleon applicants: 18% vs 6% severe crashes; 217 deaths, 3,561 injuries	GAO-12-364	B
Insurance record $0.102/mile (2024); 14 years unprofitable	ATRI 2025; Reliance Partners	B / C

Claim	Source	Class
Median nuclear verdict $51M (2024) vs $21M (2020); $750K floor since 1985	Marathon Strategies; FMCSA; Federal Register	A / C
OEM parts margin ~2.2× truck margin; razor-and-blades model	PACCAR 10-K & earnings; Rush Enterprises 8-K	A / B
Mentorship roughly doubles small-business survival (87/75; 70%)	U.S. Small Business Administration	A
1→20 trucks in 7 months; unseen fleets; $112K/$400K losses	Primary operator testimony (Michael Thomas)	E

This document synthesizes three research bodies produced in 2026 — a master structural analysis, a forensic correction re-ranking repair and maintenance as the proximate cause of failure, and a unified analysis identifying the knowledge deficit as the root cause — together with the author's operational record and the source material underlying the Heavy-Duty Repair Transparency and Fairness Act. Where earlier drafts' figures conflicted with later verification, the verified current figure has been used throughout.